Deep Learning: Final Project
Image Geolocation Challenge

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Experimental setup

I used the same country-stratified 80/20 train–validation split (9,408/2,352 images, random state 42) throughout the experiments. The backbone was ImageNet-pretrained MobileViTV2-1.0 [1]; the final 108-output model has 4,444,245 parameters. Model selection used validation median Haversine distance, with patience-four early stopping. Table 1 records the development path rather than only the successful final configuration.

Table 1: Experiment progression. "Raw → tuned" separates the trained checkpoint from validation-selected inference calibration.

#	Experiment	Median (km)	Main observation
1	Normalised latitude/longitude regression	625.8	Direct MSE regression learned a broad geographic compromise but remained inaccurate.
2	Cartesian (x, y, z) regression	Not retained	Removing longitude discontinuity did not provide a useful improvement over direct regression.
3	Country auxiliary head	516.9	Country supervision gave the shared backbone a stronger coarse geographic signal.
4	Post-hoc country-centre blend	397.5	Probability-weighted country centres were more reliable than the coordinate head.
5	Integrated country blend	373.1	Training through the blended coordinate aligned the loss with the inference calculation.
6	64 global geographic cells	357.7	Finer regions improved locality, although direct-coordinate refinement still added noise.
7	Cell-only fine-tuning and temperature	320.9 → 304.1	Removing direct coordinates helped; sharper cell probabilities improved the median but increased some large errors.
8	Country-gated country-specific cells	283.0 → 270.5	Soft country probabilities suppressed geographically inconsistent cells without hard exclusion.
9	Equal-population cell construction	278.3	More even class counts did not outperform ordinary country-specific cells; some cells lost geographic compactness.
10	Capacity-constrained K-means cells	280.8	A better balance between equal counts and compactness gave an oracle median of 80.2 km, but prediction remained the bottleneck.
11	384-pixel training	233.5 → 224.6	Higher resolution produced the largest architectural gain by preserving small visual cues.
12	512-pixel fine-tuning and multi-resolution inference	208.5 → **199.1**	Averaging 384/512 logits improved all four validation metrics without adding parameters.

What failed and what changed

Regression was the wrong abstraction. The first models asked a small network to infer two continuous numbers directly. Normalisation made optimisation stable, and Cartesian targets removed the longitude wrap-around, but neither supplied useful geographic structure. The country experiment exposed the main failure mode: correct-country predictions had much smaller errors than incorrect-country predictions. This motivated using classification probabilities as part of the coordinate calculation, rather than treating classification only as an auxiliary loss.

Increasing geographic structure improved localisation. Country centres were robust but too coarse, so I replaced them with geographic cells, following the general classification-based view of image geolocation used by PlaNet [2]. For cell j, the final model computes

$$\tilde{p}_j = \frac{p(\text{cell}_j)\, p(\text{country}(j))}{\sum_k p(\text{cell}_k)\, p(\text{country}(k))}, \qquad \hat{\mathbf{y}} = \sum_j \tilde{p}_j \mathbf{c}_j,$$

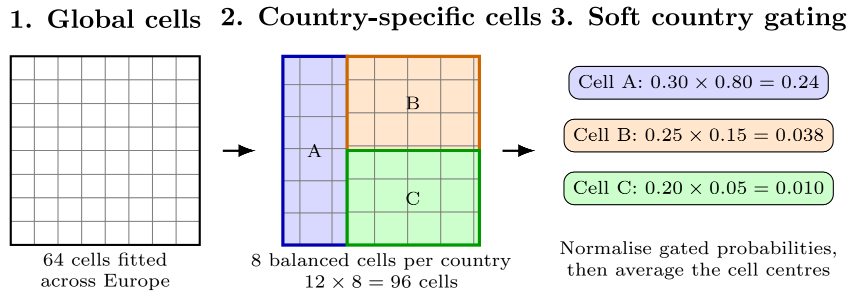

Figure 1: Schematic progression from global geographic cells to country-specific cells and soft country gating. The country prediction does not alter the cell boundaries; it increases or suppresses each cell's probability before the final coordinate is calculated.

where $\mathbf{c}_j$ is the cell centre. This soft gate was preferable to hard country selection because a wrong country prediction does not eliminate every alternative. The loss combined coordinate MSE with country and cell cross-entropy, each weighted by 0.01, so coordinate error propagated through both probability heads.

Balancing cell populations was not automatically beneficial. The first balanced partition improved class counts but produced less coherent regions and a worse score. Capacity-constrained K-means, applied separately within each country and with longitude scaled by mean latitude, produced equal-sized yet more compact cells. Its 80.2 km oracle median showed that cell resolution was no longer the main limitation: confusing cells was.

Image resolution mattered more than further cell tuning. After several cell variants produced only small changes, increasing resolution from the pretrained 256-pixel setting to 384 pixels reduced the raw median from 280.8 to 233.5 km. Fine-tuning the best checkpoint at 512 pixels reduced it to 208.5 km. This suggests that road signs, markings, poles, vegetation and architectural details had been lost during lower-resolution preprocessing.

The final inference pass evaluates the same weights at 384 and 512 pixels, averages their logits equally, and applies temperatures 0.25 (country) and 0.75 (cell). This gave a validation mean of 497.7 km, median of 199.1 km, 50.04% within 200 km and 78.61% within 750 km. The improvement from multi-resolution inference was small (201.0 to 199.1 km) but consistent across every reported metric and required no additional parameters.

Limitations and conclusion

Temperatures and resolution weights were chosen on the same validation split, so the final 199.1 km estimate may contain mild post-processing overfit. I therefore stopped after a small, coarse search rather than tuning fractional values. The mean remains much larger than the median, indicating occasional wrong-country or distant-cell failures. Nevertheless, the experiments show a clear result: under a strict parameter budget, structured region probabilities and higher-resolution inputs were substantially more effective than direct coordinate regression. The final system reduced the validation median from 625.8 to 199.1 km while keeping a single 4.44M-parameter backbone.

References

[1] S. Mehta and M. Rastegari, "Separable Self-attention for Mobile Vision Transformers," *arXiv:2206.02680*, 2022.

[2] T. Weyand, I. Kostrikov, and J. Philbin, "PlaNet—Photo Geolocation with Convolutional Neural Networks," in *European Conference on Computer Vision (ECCV)*, 2016.